

DAX Measures Explained

This document provides the DAX code and a plain-English explanation for the selected key measures used in the Hospital General Information Dashboard, presented in the requested order.

1. Total Hospitals

```
Total Hospitals = COUNTROWS(Hospital)
```

Explanation: Counts the total number of rows in the Hospital table. Each row represents one hospital, so this gives the total number of hospitals in the current filtered view.

2. % High-Performing

```
% High-Performing =  
DIVIDE(  
    [Hospitals Above Threshold],  
    [Total Hospitals],  
    0  
)
```

Explanation: Takes the count of hospitals above the rating threshold and divides it by the total number of hospitals. The result is a percentage (e.g., 0.36 = 36%). Returns 0 if Total Hospitals is zero to avoid errors.

3. Average Better %

```
Average Better % =  
DIVIDE(  
    [Safety Better %] + [Readm Better %],  
    2  
)
```

Explanation: Averages only the Safety and Readmission "Better" percentages (excludes Mortality). This is used when you want to focus specifically on these two operational domains.

4. Birthing Friendly %

```
Birthing Friendly % = DIVIDE([Birthing Friendly Hospitals], [Total Hospitals], 0)
```

Explanation: Divides the count of birthing-friendly hospitals by the total hospitals to get a percentage. The third argument (0) is a safe fallback — if Total Hospitals is zero (which would cause a divide-by-zero error), it returns 0 instead of an error.

5. Avg Overall Rating

```
Avg Overall Rating = AVERAGE(Hospital[OverallRating])
```

Explanation: Calculates the mathematical average of the OverallRating column (values 1 to 5) across all hospitals in the current filter. This gives the network-wide average star rating.

6. Overall Better %

```
Overall Better % =  
DIVIDE(  
    [Mort Better %] +  
    [Safety Better %] +  
    [Readm Better %],  
    3  
)
```

Explanation: Takes the "Better" percentages from all three domains (Mortality, Safety, Readmission), adds them together, and divides by 3 to get a simple average. For example: $(3.96\% + 19.61\% + 5.00\%) / 3 =$ approximately 9%. This gives a single number representing overall hospital quality.

7. Mort Better %

```
Mort Better % =  
DIVIDE(  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Mort",  
        PerformanceDomain[Metric] = "Better"  
    ),  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Mort",  
        PerformanceDomain[Metric] = "FacilityMeasureCount"  
    ),  
    0  
)
```

Explanation: The numerator sums all Mortality "Better" values. The denominator sums all Mortality "FacilityMeasureCount" values (the total number of facilities measured in this domain). Dividing them gives the percentage of mortality measures that scored "Better". Returns 0 if the denominator is zero.

8. Safety Better %

```
Safety Better % =  
DIVIDE(  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Safety",  
        PerformanceDomain[Metric] = "Better"  
    ),  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Safety",  
        PerformanceDomain[Metric] = "FacilityMeasureCount"  
    ),  
    0  
)
```

Explanation: Calculates the percentage of Safety measures that scored "Better". In this dashboard, this equals 19.61% — the highest "Better" percentage across all domains.

9. Readm Better %

```
Readm Better % =  
DIVIDE(  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Readm",  
        PerformanceDomain[Metric] = "Better"  
    ),  
    CALCULATE(  
        SUM(PerformanceDomain[Value]),  
        PerformanceDomain[Domain] = "Readm",  
        PerformanceDomain[Metric] = "FacilityMeasureCount"  
    ),  
    0  
)
```

Explanation: Percentage of Readmission measures that scored "Better". In this dashboard, equals 5.00%.

10. High-Performing Hospitals

```
High-Performing Hospitals =  
VAR SelectedThreshold =  
    SELECTEDVALUE('Rating Threshold'[Rating Threshold], 3)  
RETURN  
CALCULATE(  
    COUNTROWS(Hospital),  
    FILTER(  
        Hospital,  
        Hospital[OverallRating] >= SelectedThreshold  
    )  
)
```

Explanation: First, it reads the value from the Rating Threshold slicer (defaulting to 3 if nothing is selected) and stores it in a variable called SelectedThreshold. Then it counts only the hospitals whose OverallRating is greater than or equal to that threshold. So if the user sets the slider to 4, it counts only 4-star and 5-star hospitals.

11. Projected Safety Better

```
Projected Safety Better =  
[Safety Better] *  
(1 + [Safety Improvement Value])
```

Explanation: Multiplies the current Safety Better score by the improvement factor from the Safety slider. A 50% improvement (0.50) would multiply the score by 1.50.

12. Projected Readmit Better

```
Projected Readmit Better =  
[Readm Better] *  
(1 + [Readmission Improvement Value])
```

Explanation: Takes the current Readm Better score and multiplies it by (1 + improvement percentage). For example, if Readm Better is 1,330 and the slider is set to 10% (0.10), the calculation is $1,330 \times 1.10 = 1,463$. This projects what the score would look like after a theoretical improvement.

13. Best Performing Domain

```
Best Performing Domain =  
VAR Mort = [Mort Better %]  
VAR Safety = [Safety Better %]  
VAR Readm = [Readm Better %]  
  
VAR Highest =  
MAX(  
    MAX(Mort, Safety),  
    Readm  
)  
  
RETURN  
SWITCH(  
    Highest,  
    Mort, "Mortality",  
    Safety, "Safety",  
    Readm, "Readmission"  
)
```

Explanation: This measure determines which domain performs best. Step 1: It stores the "Better %" for each domain into variables (Mort, Safety, Readm). Step 2: It uses nested MAX functions to find the largest of the three values. Step 3: It uses SWITCH to compare the highest value against each variable and returns the name of the winning domain as text (e.g., "Safety"). In this dashboard, Safety wins at 19.61%.